Möbius-Admissible Completions, Short Correlations, and RH-Scale Excursions: A Literature and Research Map

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Abstract

This document is a research map for a deterministic extremal program around the Möbius summatory function. The starting space is the full ternary cube $\{-1,0,1\}^N$. Successive models impose explicit pieces of Möbius information and ask what endpoint sums and short correlations are still possible among all surviving completions. The current program has two connected parts. The first studies contraction of the adversarial endpoint envelope under arithmetic information. The second studies whether a large endpoint excursion forces an early change in the short-correlation vector relative to the true Möbius word. The document records the exact finite models, closed-form baseline results, locality barriers, short-correlation identities, the current exhaustive-computation specification, and the relevant literature. The individual ingredients belong to established areas such as multiplicative functions, discrepancy, random multiplicative functions, sign patterns, and Mertens-function theory. The specific finite completion and short-correlation questions are therefore stated with explicit rule systems and with novelty claims kept separate from the mathematical definitions.

1 The organizing problem

Let

$$\Omega_N^{(0)} = \{-1,0,1\}^N, \qquad |\Omega_N^{(0)}| = 3^N,$$

and let $\Omega_N^{(j)} \subseteq \Omega_N^{(0)}$ be the set of words surviving a specified collection $\mathcal{I}_j$ of rigorous arithmetic constraints. For $\omega = (\omega_1, \ldots, \omega_N)$ put

$$S_\omega(n) = \sum_{k \le n} \omega_k, \qquad B_N^{(j)} = \max_{\omega \in \Omega_N^{(j)}} |S_\omega(N)|.$$

The quantity

$$E_N^{(j)} = \frac{\log B_N^{(j)}}{\log N} - \frac{1}{2}$$

was used in the first finite tables, but it is not a reliable finite-scale estimate of the growth exponent. If

$$B_N = cN^\theta,$$

then

$$E_N = \theta - \frac{1}{2} + \frac{\log c}{\log N},$$

so a small constant c can make a linear envelope look close to square-root scale at small N. The main diagnostics are therefore

$$R_N^{(j)} = \frac{B_N^{(j)}}{N}$$

and the local logarithmic slope

$$\theta_N^{(a,j)} = \frac{\log B_{aN}^{(j)} - \log B_N^{(j)}}{\log a},$$

with $E_N^{(j)}$ retained only as a historical finite statistic. The central deterministic question is not whether a large excursion is improbable, but whether any admissible completion producing it remains.

A useful filtration has the schematic form

$$\Omega_N^{(0)} \supseteq \Omega_N^{(\mathrm{sqf})} \supseteq \Omega_N^{(\mathrm{mult})} \supseteq \Omega_N^{(P_r)} \supseteq \Omega_N^{(\mathrm{id})} \supseteq \cdots,$$

where the precise information content of every layer must be stated independently of the actual unknown values one is attempting to bound. Imposing the complete definition of Möbius together with complete factorization reduces the space to the unique true Möbius word and is therefore an endpoint check, not a nontrivial layer of the program.

2 RHC, RHF$_\delta$, and RHFl

For a finite word, define its endpoint excess exponent by

$$e_N(\omega) = \frac{\log |S_\omega(N)|}{\log N} - \frac{1}{2},$$

when $S_\omega(N) \neq 0$.

For every $\delta \geq 0$, define

$$\mathrm{RHF}_{N,\delta} = \{\omega : |S_\omega(N)| > N^{1/2+\delta}\}.$$

The square-root class is the case $\delta = 0$:

$$\mathrm{RHC}_N = \mathrm{RHF}_{N,0} = \{\omega : |S_\omega(N)| > \sqrt{N}\}.$$

Thus RHC is not the case $\delta = 1/2$. The latter would require $|S_\omega(N)| > N$, which is impossible for a word with $|\omega_n| \leq 1$.

A finite RHC event is not incompatible with RH. The disproved Mertens conjecture asserted the much stronger unit-constant bound $|M(N)| \leq \sqrt{N}$ for every N. For fixed $\delta > 0$, a single finite event in $\mathrm{RHF}_{N,\delta}$ also does not falsify RH. The RH-relevant asymptotic statement is the exclusion of a fixed positive excess exponent on unbounded scales.

An RHFl event is different. It is a finite violation of an explicit proved consequence of RH. If a theorem gives

$$\mathrm{RH} \Longrightarrow |M(x)| \leq H_{\mathrm{RH}}(x)$$

in a stated range, then a value larger than $H_{\mathrm{RH}}(x)$ is locally incompatible with RH in that range. This is kept separate from the model classes $\mathrm{RHF}_{N,\delta}$.

3 Classical Mertens-function background

RH is equivalent to

$$M(x) = O_\varepsilon(x^{1/2+\varepsilon}) \qquad \text{for every } \varepsilon > 0.$$

This is an asymptotic statement with an ε-dependent constant; therefore no unit-constant inequality $|M(N)| \le N^{1/2+\varepsilon}$ at one finite N is an RH consequence. Ng studies the conditional distribution and extreme-value behavior of $M(x)$ under RH together with additional hypotheses, including a limiting distribution for the normalized summatory function and a heuristic extreme scale involving $\sqrt{x}(\log\log\log x)^{5/4}$ [23]. Explicit conditional estimates, such as those of Simonič, are useful for defining a genuine RHFl boundary but only in their stated ranges [24].

4 Discrepancy and adversarial multiplicativity

The discrepancy literature is essential because it is deterministic and adversarial rather than probabilistic. Tao proved the Erdős discrepancy problem: every $\{\pm 1\}$ sequence has unbounded discrepancy on homogeneous arithmetic progressions [1]. For a completely multiplicative $\{\pm 1\}$-valued sequence, discrepancy on a homogeneous progression factors through the ordinary partial sum, so the multiplicative specialization asks whether ordinary partial sums can remain bounded. Earlier and related work studies multiplicative functions with bounded or forced-unbounded partial sums [2, 3].

The conceptual lesson for the present program is sharp: a severe reduction in the number of admissible multiplicative configurations need not imply a comparable reduction in the extremal partial sum. Discrepancy results force excursions under structural constraints; the desired Möbius-side counterpart would be an upper-bound or contraction theorem showing that sufficiently strong Möbius-specific constraints forbid excursions of fixed power above $N^{1/2}$.

5 Structure of multiplicative functions with small sums

Koukoulopoulos and Soundararajan characterize strong structural consequences of unusually small partial sums of multiplicative functions [4]. This literature is close to the contrapositive viewpoint of the present program: instead of asking what structure follows from small sums, one asks how much a prescribed structure restricts the largest possible sum. Any claimed novelty must therefore be phrased at the level of *partial-information completion* and the quantitative contraction of its extremal envelope, not merely as an extremal problem over multiplicative functions.

6 Möbius-like multiplicative functions on squarefree and k-free support

Aymone gives explicit multiplicative functions supported on the squarefree integers, with prime values $f(p) = \pm 1$, whose summatory functions satisfy $M_f(x) = o(\sqrt{x})$; under RH the construction can go beyond square-root cancellation [9]. Aymone, Bueno, and Medeiros extend this to multiplicative functions supported on the k-free integers, constructing examples with partial sums $o(x^{1/k})$ and stronger conditional exponents under GRH [10]. Aymone also studies the Erdős discrepancy problem on squarefree and cubefree supports, proving unbounded partial sums for natural multiplicative sequences built from these supports [11].

These papers are especially important for the present program because they show that the layer "squarefree support + multiplicativity + free prime signs" already belongs to an established extremal literature. Therefore a new contribution cannot consist merely of optimizing over prime signs on squarefree support. Any distinct object must use genuinely Möbius-specific information closure or quantify the marginal effect of information that these model classes do not encode.

7 Random multiplicative functions as a scale reference

Wintner's Rademacher random multiplicative model is a classical probabilistic model for Möbius. Modern work gives a detailed picture of its large fluctuations. Harper proves almost sure arbitrarily large fluctuations at scale

$$\sqrt{x}(\log\log x)^{1/4+o(1)},$$

showing in particular why crossing $\sqrt{x}$ is not itself an RH-false phenomenon [7]. Harper's high-moment work gives large-deviation information for Rademacher and Steinhaus random multiplicative sums [6]. Recent upper-bound work determines the matching logarithmic scale in these models [8].

These results are not evidence that RH is "probable." Their role here is diagnostic: they identify a natural multiplicative fluctuation scale of the form $x^{1/2+o(1)}$ and therefore provide a reference against which deterministic RHC and RHF envelopes can be interpreted.

8 Biased random multiplicative functions and RH-equivalent model statements

Aymone and Sidoravicius study random multiplicative functions supported on the squarefree integers whose prime values are biased toward the classical Möbius function [12]. Their results include necessary and sufficient conditions for power-saving partial sums in biased models and an RH-equivalent statement formulated through a family of weakly biased random multiplicative functions. This is a particularly close antecedent to any attempt to interpret progressive Möbius information probabilistically.

For the present deterministic program the distinction is decisive: conditioning or biasing a probability law on prime values does not eliminate every exceptional completion. Nevertheless, this literature supplies both warning and calibration: a filtration whose only new ingredient is a growing bias toward $\mu(p) = -1$ is unlikely to constitute a distinct research problem.

9 Local sign patterns and realizability

Matomäki, Radziwiłł, and Tao prove positive-density occurrence results for local sign patterns of Liouville and for the nine possible pairs $(\mu(n), \mu(n+1))$ [5]. This addresses local realizability of Möbius patterns, whereas the present completion problem concerns global prefixes $(1, \dots, N)$ subject to simultaneous arithmetic constraints. Nevertheless, sign-pattern theorems can become genuine constraints when a filtration contains translated windows or local-pattern requirements; they must not be reintroduced as if they were new combinatorial observations.

10 Partial multiplicativity and prescribed prime information

Vandehey explicitly studies weakened multiplicativity in which multiplicative relations are imposed only for primes in a finite prescribed set [3]. This is a direct antecedent of any layer parameterized merely by a finite set of known primes. Consequently, "reveal the first r primes and optimize" is not by itself a sufficiently distinct research object.

The potentially different object is instead the *closure* of partial Möbius information: after revealing a specified information set, include every value, zero, sign relation, residual primality conclusion, and multiplicative identity that follows logically from it, and optimize over all completions consistent with that closure. The mathematical object is therefore a constraint-closure operator rather than a list of fixed prime signs.

11 Pretentiousness and inverse information from partial sums

The pretentious approach of Granville and Soundararajan organizes multiplicative functions through a quantitative distance measuring how closely one multiplicative function imitates another [13]. Koukoulopoulos–Soundararajan and subsequent work of Sachpazis develop inverse principles in which unusually small partial sums force strong average structure on prime values [4, 14]. This literature is relevant to the research map in two directions. First, it shows that partial-sum behavior itself carries information about the underlying prime data. Second, it provides candidate analytic constraints that could eventually replace brute-force propagation by structural inequalities.

A future completion theorem should therefore be compared not only with discrepancy theory but also with these inverse theorems: if a proposed "new" constraint is already a consequence of pretentious distance or an inverse theorem for small sums, the literature result should be used directly.

12 Computational Mertens benchmarks

Direct computation of the true Mertens function is not the objective of the completion program, but it supplies indispensable controls. Hurst computed all values through 10^{16} and improved numerical bounds associated with the disproved Mertens conjecture [21]. A 2026 implementation computes isolated values through 10^{25}, including $M(10^{24})$ and $M(10^{25})$, while retaining the classical $O(x^{2/3+\varepsilon})$ isolated-value framework [22]. These computations make clear that finite verification of the true word can be pushed vastly farther than exhaustive enumeration of completion spaces. Consequently, computational novelty must come from solving the adversarial completion problem, not from tabulating $M(N)$ itself.

13 Chudakov-type classification and bounded partial sums

A further novelty boundary comes from classification theorems for multiplicative functions with bounded partial sums. Klurman proves Chudakov's conjecture in a finite-range setting: under the appropriate hypotheses, a completely multiplicative function with bounded partial sums is forced into character-like structure [15]. Klurman, Mangerel, Pohoata, and Teräväinen subsequently develop sieve-theoretic tools for multiplicative functions whose partial sums stay close to their mean and obtain, among other consequences, strong rigidity statements for bounded partial sums [16]. Aymone gives additional classifications and constructions for complex-valued multiplicative functions with bounded partial sums [17].

These results occupy a large part of the naive extremal question "which multiplicative assignments have bounded sums?" The present program must therefore retain its finite-information aspect: the variable object is not a completed multiplicative function selected from a broad class, but the adversarial envelope of all completions consistent with a declared, incomplete Möbius information state.

14 Finite prime modifications and stability of partial sums

Aymone studies completely multiplicative functions obtained by modifying a primitive Dirichlet character at a finite set of primes, proving Ω-bounds whose scale depends explicitly on the number of modified primes [18]. This is very close to any experiment in which finitely many prime values are altered while all remaining prime data are frozen. It shows that the location and number of modified prime generators can leave a quantitative signature in the partial sums.

Consequently, a layer obtained merely by "freeing" or "fixing" finitely many prime signs is not enough to define a new object. The potentially distinct ingredient is again the closure map: partial information must generate all deterministic Möbius consequences before the adversarial maximum is taken.

15 Further Möbius-like extremal constructions

Liu studies classes of multiplicative functions resembling Möbius and proves complementary O- and Ω-results for their summatory functions [19]. Together with the squarefree and k-free constructions already reviewed, this reinforces the point that extremizing partial sums over Möbius-like multiplicative families is itself established territory. The completion program should use these papers as test families: any proposed envelope theorem should be checked against known constructions that deliberately produce unusually small or unusually large sums.

16 Exponential-sum rigidity with multiplicative coefficients

Recent work of Bazin studies bounded exponential sums with multiplicative coefficients and develops structural consequences from boundedness, extending ideas related to Chudakov-type rigidity to more general oscillatory observables [20]. Although the observable differs from the ordinary Mertens sum, this literature is relevant methodologically: it supplies additional examples where a global bound forces pretentious or character-like structure. Such inverse mechanisms may eventually be convertible into deterministic pruning rules for a completion algorithm.

17 Targeted search for completion, query, and certificate formulations

A dedicated search was carried out for formulations under terms such as *partial specification of multiplicative functions*, *extension/completion of multiplicative functions*, *reconstruction of Möbius from partial values*, *query complexity of multiplicative functions*, and *certificate complexity for Möbius*. The closest primary-source results located were partial multiplicativity on prescribed prime sets, inverse theorems from small partial sums, rigidity/classification results for bounded sums, finite modifications of characters, and probabilistic conditioning on prime values. No primary-source

paper was located that defines the finite adversarial completion functional

$$\mathfrak{B}_N(\mathcal{I}) = \max_{\omega \in \mathcal{C}_N(\mathcal{I})} \left| \sum_{n \le N} \omega_n \right|$$

for a progressively closed body of Möbius-specific information and then studies its contraction as an information-complexity problem. This is a negative search result, not a proof of novelty, and should be updated as the bibliography grows.

18 Novelty-threat matrix

The following table is intended as a guardrail against rediscovery.

Candidate idea	Closest established literature	Status	What would still be different
Free signs on squarefree support	Aymone; Aymone–Bueno–Medeiros; Liu	Occupied	Möbius-specific information closure
Finite set of prescribed prime values	Vandehey; modified-character literature	Occupied	Optimize after full logical closure, not after raw prescriptions
Bounded/unbounded multiplicative partial sums	Tao; Klurman; KMPT; Aymone	Occupied	RH-scale upper envelope under partial information
Probabilistic progressive prime information	Aymone–Sidoravicius; random multiplicative literature	Occupied	Deterministic elimination of every surviving completion
Inverse structure from small sums	Koukoulopoulos–Soundararajan; Sachpazis; Bazin	Occupied	Forward contraction from declared finite information
Count of surviving words	Elementary combinatorics / CSP viewpoint	Not sufficient	Coupled with arithmetic closure and extremal envelope
Adversarial Möbius completion envelope	No exact antecedent located in targeted search	Open search status	Formalize closure, information cost, and N-scaling
Minimal information killing every RHF_δ completion	No exact antecedent located in targeted search	Open search status	Query/certificate complexity of RH-critical exclusion

19 The completion operator

Let $\mathcal{I}$ be a finite body of arithmetic information available at scale N. Define

$$\mathcal{C}_N(\mathcal{I}) = \{\omega \in \{-1, 0, 1\}^N : \omega \text{ satisfies every consequence of } \mathcal{I} \text{ included in the model}\}.$$

The fundamental extremal functional is

$$\mathfrak{B}_N(\mathcal{I}) = \max_{\omega \in \mathcal{C}_N(\mathcal{I})} \left| \sum_{n \le N} \omega_n \right|.$$

For nested information sets $\mathcal{I} \subseteq \mathcal{J}$,

$$\mathcal{C}_N(\mathcal{J}) \subseteq \mathcal{C}_N(\mathcal{I}), \qquad \mathfrak{B}_N(\mathcal{J}) \le \mathfrak{B}_N(\mathcal{I}).$$

The contraction produced by adding information is

$$\Delta_{\mathcal{I}\to\mathcal{J}}(N) = \mathfrak{B}_N(\mathcal{I}) - \mathfrak{B}_N(\mathcal{J}),$$

or, on the exponent scale,

$$\Delta E_{\mathcal{I}\to\mathcal{J}}(N) = \frac{\log \mathfrak{B}_N(\mathcal{I}) - \log \mathfrak{B}_N(\mathcal{J})}{\log N}.$$

This distinguishes information that merely reduces the number of completions from information that actually contracts the dangerous extremal envelope.

20 Constraint families to be tested

The following layers should be treated as separate mathematical experiments.

1. Full ternary space $\{-1, 0, 1\}^N$.
2. Möbius support information: $\mu(1) = 1$, squarefree/non-squarefree constraints.
3. Multiplicative compatibility with prime signs still free.
4. Finite-prime partial multiplicativity, explicitly compared with the Vandehey-type antecedent.
5. Sieve information available below a moving threshold, especially $p \le \sqrt{N}$.
6. Residual closure by the first r primes: factor extraction plus every forced residual-primality conclusion licensed by the chosen closure rule.
7. Möbius identities that couple many coordinates without revealing them individually.
8. Prefix or endpoint information, when it is part of the experiment, with its information cost recorded separately.
9. Proven local pattern/correlation constraints when they can be encoded deterministically.
10. The full Möbius word only as a terminal control, never as evidence that an intermediate filtration contracts.

21 Quantities to record at every layer

For every arithmetic layer (N, j), the basic record contains

$$|\Omega_N^{(j)}|, \qquad B_N^{(j)}, \qquad R_N^{(j)} = \frac{B_N^{(j)}}{N}, \qquad \theta_N^{(a,j)},$$

with $E_N^{(j)}$ retained only for comparison with the original finite tables. For a declared grid of $\delta \ge 0$, record the exact counts

$$\#\{\omega \in \Omega_N^{(j)} : |S_\omega(N)| > N^{1/2+\delta}\}.$$

The case $\delta = 0$ is the RHC count. Counts are descriptive; for an adversarial upper-bound question, a single surviving completion is enough to prevent a universal conclusion.

For the short-correlation experiment, the record also contains the first witness lag h_*, the full survivor curve $R_{N,\delta}(H)$, witness histograms, quantiles of h_*, record-breaking adversaries, and the fraction of the total correlation excess that appears in short lags. These quantities are defined below.

22 Literature-to-constraint map

Literature	Object / mode	Direct use here	Main limitation
Tao [1]	Discrepancy of ± 1 sequences; deterministic	Forced excursions; adversarial benchmark	Not a square-root/RH-scale upper bound
Vandehey [3]	Partial multiplicativity on a finite prime set; deterministic	Antecedent for partial-prime layers	Does not build Möbius-specific closure
Koukoulopoulos–Soundararajan [4]	Multiplicative functions with small partial sums; structural	Benchmark for extremal sums and inverse constraints	Reverse direction from completion contraction
Matomäki–Radziwiłł–Tao [5]	Local Liouville/Möbius sign patterns; deterministic	Realizability and local constraints	Translated local windows, not global prefix completions
Harper [7, 6]	Random multiplicative partial sums; probabilistic	Natural $x^{1/2+o(1)}$ scale and large deviations	Cannot eliminate exceptional deterministic completions
Durkan–Pearce-Crump [8]	Sharp a.s. upper scale in random models	Reference scale for RHC	Random model, not Möbius
Aymone [9]	Squarefree-supported Möbius-like functions; constructive	Shows free prime-sign squarefree layer is established territory	No progressive Möbius-specific closure
Aymone–Bueno–Medeiros [10]	k-free supported multiplicative functions; constructive	Extremal cancellation benchmark on restricted support	Different information model
Aymone–Sidoravicius [12]	Biased random Möbius-like functions	Warning for progressive prime bias; RH-equivalent model results	Exceptional deterministic completions remain
Granville–Soundararajan [13]	Pretentious distance; analytic	Structural metric for prime information	Not a finite completion envelope
Sachpazis [14]	Inverse structure from small partial sums	Candidate source of global constraints	Reverse/inverse direction
Klurman [15]	Chudakov-type rigidity; completely multiplicative functions	Classification benchmark for bounded sums	Completed functions, not partial-information envelopes
Klurman–Mangerel–Pohoata–Teräväinen [16]	Sieve/inverse structure for multiplicative partial sums	Rigidity and potential pruning inequalities	Does not optimize finite Möbius completions
Aymone [18]	Finite modifications of Dirichlet characters	Quantifies effects of finitely changed prime values	Remaining prime data are not an unknown completion state

Literature	Object / mode	Direct use here	Main limitation
Liu [19]	Möbius-like multiplicative functions; O/Ω bounds	Stress-test family for envelope claims	No progressive information filtration
Bazin [20]	Bounded exponential sums with multiplicative coefficients	Further inverse/rigidity mechanisms	Different observable and no Möbius completion functional
Hurst [21, 22]	Direct computation of $M(x)$	Ground truth and scaling benchmark	Computes the unique true word, not admissible completions
Ng [23]	Conditional distribution of $M(x)$	RH-compatible extreme scale	Uses RH plus additional hypotheses
Simonič [24]	Explicit conditional bounds for $M(x)$	Genuine RHFl boundary where applicable	Very large validity thresholds

23 Gap analysis

The following claims are *not* candidates for novelty by themselves: enumerating ternary words; imposing squarefree support; parameterizing multiplicative functions by prime values; conditioning or prescribing finitely many prime values; asking whether multiplicative partial sums are bounded; studying large deviations of random multiplicative sums; or proving that particular local Möbius sign patterns occur.

A potentially distinct problem remains if all of the following are essential:

1. the objects are finite Möbius-admissible *completions*, not a probability model;
2. information is represented by an explicit closure operator that propagates every permitted deterministic consequence;
3. the objective is the adversarial maximum $\mathfrak{B}_N(\mathcal{I})$ over all surviving completions;
4. the information content is varied systematically, allowing marginal contraction to be assigned to particular arithmetic constraints;
5. the target scale is the disappearance of every fixed-power excess $N^{1/2+\delta}$, not merely bounded discrepancy or small average behavior.

A literature claim of novelty should be made only after a dedicated search for extremal completion, partial specification, reconstruction/query complexity, and constraint-satisfaction formulations for multiplicative functions.

24 Restricted-information RH target

Every valid information layer contains the true Möbius prefix. Therefore

$$B_N(\mathcal{I}) \geq |M(N)|.$$

Any explicitly defined information family satisfying

$$B_N(\mathcal{I}(N)) \leq N^{1/2+o(1)}$$

immediately implies RH. If arbitrary information is allowed, including full revelation of the Möbius word, the mere existence of such an information family is tautological once RH is assumed and is not a useful formulation. The rule class must therefore be fixed in advance and must exclude terminal encodings of the target word. For a fixed restricted class $\mathfrak{I}$, one may ask whether a genuine equivalence of the form

$$\mathrm{RH} \iff \exists \mathcal{I}_N \in \mathfrak{I} : \ B_N(\mathcal{I}_N) \le N^{1/2+o(1)}$$

can be obtained. The forward direction is the difficult one because it requires a natural restricted family rather than arbitrary revelation.

25 Constraint families kept in parallel

The current map uses locality as an additional classification, not as a replacement for the arithmetic routes.

1. **Arithmetic/divisorial:** squarefree support, multiplicativity, prime-sign revelation without residual propagation, residual closure P_r, Möbius identities, reconstruction rules, and broader convolution or bilinear identities.

2. **Translational:** sign patterns, shifted products, local blocks, short correlations, and aggregate translated statistics.

3. **Global analytic:** Dirichlet functionals, exponential sums, smooth transforms, and other rows that touch a positive fraction of the coordinates.

4. **Hybrid:** combinations of arithmetic, translational, and global constraints.

A no-go theorem for one locality class is recorded with its hypotheses and does not remove the other classes from the program.

26 Research questions

The current program keeps several routes in parallel.

1. Which arithmetic constraints reduce only the number of completions, and which reduce the extremal envelope B_N?

2. How do the two partial-prime-information models—raw prime-sign revelation and residual closure P_r—differ in their surviving extremal envelopes?

3. Which arithmetic constraints remain strictly divisor-local, and which aggregated arithmetic identities transport information across a positive fraction of the coordinates?

4. For the short-correlation model, how does

   $$H_{\max,\delta}(N)$$

   grow with N and δ?

5. Does a fixed positive power excursion force a witness at a sublinear lag, for example $O(\log N)$, $O(N^{\alpha})$ with $\alpha < 1$, or another scale?

6. Is the true Möbius word central, extremal, or Pareto-minimal in natural finite correlation geometries of the admissible cube?

7. Which of these finite observations can be converted into a theorem that excludes every fixed-power excursion of the true Möbius word?

8. For a fixed non-tautological rule class $\mathcal{I}$, can one characterize the information needed to force an envelope of size $N^{1/2+o(1)}$ without allowing the full Möbius word as input data?

27 Computational protocol

The production experiment discussed here uses the squarefree multiplicative family with the sign of 2 fixed:

$$x_1 = 1, \qquad x_n = 0 \quad (n \text{ not squarefree}),$$

and, for squarefree n,

$$x_n = \prod_{p \mid n} \sigma_p, \qquad \sigma_2 = -1, \qquad \sigma_p \in \{-1, +1\} \quad (p \geq 3).$$

There are exactly

$$2^{\pi(N)-1}$$

states. Bit value zero represents $\sigma_p = -1$, so the all-zero mask is the true Möbius word.

The optimized engine is written in C++ and enumerates the binary cube in Gray order. If the sign at a free prime p is flipped, only the squarefree positions divisible by p change sign. Let

$$V_p = \{n \leq N : p \mid n,\ n \text{ squarefree}\}.$$

Then the endpoint sum is updated by

$$S' = S - 2 \sum_{n \in V_p} x_n.$$

For

$$C_h(x; N) = \sum_{n \leq N-h} x_n x_{n+h},$$

define the affected pairs

$$E_{p,h} = \{n \leq N - h :\ p \mid n \text{ xor } p \mid n + h,\ x_n x_{n+h} \neq 0\}.$$

Before the sign flip is applied to the stored word,

$$C_h' = C_h - 2 \sum_{n \in E_{p,h}} x_n x_{n+h}.$$

The implementation must test the update order explicitly because changing the stored word before evaluating the old products reverses the sign convention.

The Gray bits are assigned by measured update cost rather than by prime size alone. A useful cost is

$$\kappa(p) = |V_p| + \sum_{h \leq H_0} |E_{p,h}|,$$

and the variables with the smallest $\kappa(p)$ receive the highest flip frequency.

The lag search is adaptive. A small initial block, for example $1 \le h \le 8$, is maintained incrementally in the hot loop. Excursion states that survive this block are examined in later blocks

$$9{:}16, \quad 17{:}32, \quad 33{:}64, \ldots$$

until a witness is found or $h = N-1$ is reached. The numbers $8, 16, 32, \ldots$ are computational block boundaries, not mathematical hypotheses. In particular, the earlier observation $h \le 9$ is not fixed as a rule.

One enumeration is used for all excursion thresholds. For each δ define the integer threshold

$$T_\delta(N) = \left\lfloor N^{1/2+\delta} \right\rfloor + 1.$$

The state belongs to $\mathrm{RHF}_{N,\delta}$ exactly when $|S| \ge T_\delta(N)$. For $\delta = 0$, the code can use the exact integer condition

$$S^2 > N.$$

It is also useful to record

$$\delta_{\mathrm{eff}}(x; N) = \frac{\log |S_x(N)|}{\log N} - \frac{1}{2},$$

so one state can be assigned to every threshold it exceeds without rerunning the enumeration.

Two independent engines are required. The reference engine reconstructs each word from its bit mask and computes S and all required C_h directly. The production engine uses the incremental Gray updates. For every state and every $N \le 25$, the two engines must agree exactly on the word, S, each C_h, excursion classification, h_*, survivor counts, and witness histograms. The known $N = 25$ endpoint benchmark is mandatory:

$$2^8 = 256 \text{ states}, \qquad \#\mathrm{RHC}_{25} = 8,$$

and

S	-6	-4	-2	0	2	4	6
#	4	24	60	80	60	24	4.

A mismatch is a fatal validation error. Larger runs use deterministic sampled recomputation checks in addition to the exact small-N validation.

The code stores aggregate statistics during the exhaustive pass and keeps only a configurable number of the hardest states. Each retained state includes the bit mask, endpoint sum, first witness lag, witness excess, correlation vector up to the required depth, and the complete ternary word. The null-model analysis is a separate layer and may use sampling; the exhaustive existence and non-existence results do not.

The exponential state count, not N itself, determines the brute-force limit. For example,

$$2^{26} = 67{,}108{,}864, \qquad 2^{30} = 1{,}073{,}741{,}824, \qquad 2^{32} = 4{,}294{,}967{,}296.$$

The engine must report states per second, thread scaling, memory use, and an ETA before a large run. A later implementation may process several endpoints with the same free-prime set in one cube traversal, but the first exact implementation may use one N per run.

28 Decision criterion for continuing the program

The baseline arithmetic layers are judged by the growth of B_N, not by the small-N value of E_N. A layer whose local slope stays near one has not produced a power contraction even if its multiplicative constant is small.

The short-correlation route is judged differently. For fixed $\delta > 0$, the central quantity is the growth of $H_{\max,\delta}(N)$ and of its high quantiles. A result with $H_{\max,\delta}(N) = o(N)$ would localize every fixed-power excursion to a sublinear lag range. A result with $H_{\max,\delta}(N) \asymp N$ would give no substantial localization beyond the global correlation identity. The case $\delta = 0$ is retained as a diagnostic baseline but cannot define a universal theorem for the true Möbius word because the bound $|M(N)| \le \sqrt{N}$ is false in general.

29 Working checklist

Before adding a new layer R:

1. identify the theorem or elementary arithmetic fact that licenses R;
2. search whether the same partial-information model has already been studied;
3. specify exactly what information R reveals and what it does not reveal;
4. compute the closure of R under the rules already admitted;
5. record the change in $|\Omega_N|$, B_N, B_N/N, local log–log slopes, RHC counts, and RHF_δ counts; retain E_N only as an archival finite-size diagnostic;
6. distinguish a reduction in multiplicity from a reduction in the extremal envelope;
7. attempt an analytic explanation of every persistent contraction point.

30 Closed forms for the first arithmetic layers

Let

$$Q(N) = \#\{n \le N : n \text{ is squarefree}\},$$

and let $Q_{\mathrm{odd}}(N)$ and $Q_{\mathrm{even}}(N)$ denote the odd and even squarefree counts. The first baseline layers are completely explicit.

For unrestricted squarefree signs,

$$\boxed{B_N^{\mathrm{sqf}} = Q(N)}.$$

After imposing multiplicativity while leaving all prime signs free, the same maximum remains attainable:

$$\boxed{B_N^{\mathrm{mult}} = Q(N)}.$$

Indeed, choosing every prime sign $+1$ makes every nonzero squarefree term positive. Multiplicativity can therefore reduce the number of admissible configurations by many orders of magnitude while producing no contraction at all in the endpoint envelope.

Since

$$Q(N) \sim \frac{6}{\pi^2} N,$$

both layers have asymptotic exponent one.

Now fix $\sigma_2 = -1$ and leave all odd-prime signs free. Pair each odd squarefree $m \le N/2$ with $2m$. Their contributions cancel exactly. Hence

$$S_N = \sum_{\substack{N/2<m\le N \\ m \text{ odd,squarefree}}} f(m),$$

and therefore

$$\boxed{B_N^{\mu(2)} = Q_{\text{odd}}(N) - Q_{\text{odd}}(N/2) = Q_{\text{odd}}(N) - Q_{\text{even}}(N)}.$$

The equality with $Q_{\text{odd}}(N) - Q_{\text{even}}(N)$ follows because every even squarefree integer up to N is uniquely $2m$ with m odd squarefree and $m \le N/2$. Since

$$Q_{\text{odd}}(N) \sim \frac{4}{\pi^2}N, \qquad Q_{\text{even}}(N) \sim \frac{2}{\pi^2}N,$$

we obtain

$$\boxed{B_N^{\mu(2)} \sim \frac{2}{\pi^2}N}.$$

Thus the transition from the squarefree/multiplicative envelope to the fixed-$\mu(2)$ envelope changes the leading constant from $6/\pi^2$ to $2/\pi^2$, an asymptotic factor of exactly three, but it does not change the exponent.

At $N = 25$ this is the finite drop

$$16 \longrightarrow 6.$$

It is a genuine contraction of the maximum, but it is a constant-factor contraction rather than a power contraction.

30.1 Why the finite effective exponent is misleading

For an envelope of the form

$$B_N = cN^\theta,$$

the finite statistic

$$E_N = \frac{\log B_N}{\log N} - \frac{1}{2}$$

satisfies

$$\boxed{E_N = \theta - \frac{1}{2} + \frac{\log c}{\log N}}.$$

The last term is a finite-size bias. At small N it can be as large as the full exponent range one is trying to distinguish. For the fixed-$\mu(2)$ layer, $c \approx 2/\pi^2$, so the small value of E_{25} does not indicate approach to square-root scaling. In fact

$$\boxed{E_N \longrightarrow \frac{1}{2}}$$

for the unrestricted ternary, squarefree, multiplicative, and fixed-$\mu(2)$ baseline layers.

The primary finite diagnostics are therefore the normalized envelope

$$\boxed{R_N = \frac{B_N}{N}}$$

and a local log–log slope, for example

$$\boxed{\theta_N^{(a)} = \frac{\log B_{aN} - \log B_N}{\log a}}.$$

If $B_N \sim cN^\theta$, the multiplicative constant cancels and $\theta_N^{(a)} \to \theta$.

For the fixed-$\mu(2)$ layer the exact endpoint maxima and the decade slopes are

N	$B_N^{\mu(2)}$	$B_N^{\mu(2)}/N$	$\dfrac{\log B_{10N}^{\mu(2)} - \log B_N^{\mu(2)}}{\log 10}$
10^2	21	0.210000	0.978811
10^3	200	0.200000	1.006252
10^4	2029	0.202900	0.999357
10^5	20260	0.202600	1.000099
10^6	202646	0.202646	$--$

The normalized envelope stabilizes near $2/\pi^2 \approx 0.202642$ and the local slope stabilizes near one. This is the appropriate diagnosis: the layer is linear.

30.2 Status of the finite benchmark

The five finite benchmark tables below remain arithmetically valid. Their counts and endpoint maxima are not superseded by the asymptotic analysis. What changes is the interpretation of the columns involving E_N: at $N \le 25$, E_N mainly reflects the multiplicative constant and should not be used as evidence of exponent contraction. The tables are therefore retained as exact finite combinatorial data, while B_N/N and local slopes are used for scaling conclusions.

31 Walsh representation of the fixed-σ_2 endpoint

In the fixed-σ_2 model the endpoint is a multilinear polynomial in the free odd-prime signs. At $N = 25$ it is exactly

$$S_{25} = \sigma_{13} + \sigma_{17} + \sigma_{19} + \sigma_{23} + \sigma_3\sigma_5 + \sigma_3\sigma_7.$$

The variable σ_{11} is silent in the endpoint. The eight RHC states are exactly the states with $|S_{25}| = 6$. Writing $\tau \in \{-1, +1\}$, their signs satisfy

$$\sigma_{13} = \sigma_{17} = \sigma_{19} = \sigma_{23} = \tau, \qquad \sigma_5 = \sigma_7 = \tau\sigma_3,$$

while σ_{11} remains free. This gives $2 \cdot 2 \cdot 2 = 8$ states.

For $N \le 25$, the endpoint monomials are independent over $\mathbb{F}_2$, so the endpoint distribution is the corresponding binomial distribution with multiplicity. The first dependence in this family occurs at $N = 65$, where the four squarefree terms $33, 39, 55, 65$ satisfy

$$f(33)f(39)f(55)f(65) = 1.$$

This dependence changes the distribution and the multiplicity of extremizers, but it does not reduce the endpoint envelope while the all-positive choice of the active monomials remains admissible.

32 Two different partial-prime-information models

Two prime-information models are kept separate because they have different closure rules.

Model A: prime signs without residual propagation. Revealed prime signs are fixed, unrevealed prime signs remain independent, and no primality conclusion is inferred from a residual factor. At $N = 25$, the exact endpoint data are

revealed signs	# states	B_{25}	#RHC	$\#\mathrm{RHF}_{25,0.10}$
$\{2\}$	256	6	8	0
$\{2,3\}$	128	6	4	0
$\{2,3,5\}$	64	6	2	0.

Thus fixing more true prime signs reduces multiplicity here without reducing the endpoint envelope below 6.

If k unrevealed primes lie in $(N/2, N]$, each occurs only at its own coordinate. After fixing the remaining variables, the endpoint has the form

$$S_N = C + \sum_{p \in P_{\mathrm{free}}} \sigma_p.$$

Hence

$$\max |S_N| = |C| + k \geq k.$$

Starting from the true Möbius word and changing those k prime signs from -1 to $+1$ also gives

$$B_N \geq 2k - |M(N)|.$$

If all primes in $(N/2, N]$ are free, then

$$k = \pi(N) - \pi(N/2) \sim \frac{N}{2 \log N},$$

which gives an $N^{1-o(1)}$ obstruction to a square-root-scale envelope.

Model B: residual closure P_r. Let $P_r = \{p_1, \ldots, p_r\}$ be the revealed primes. Strip the powers of these primes from n. A repeated known prime factor forces zero. A residual equal to one fixes the sign. If the remaining residual q has no factor in P_r and satisfies $q \leq p_r^2$, it is forced to be prime. Larger residual cores remain shared variables.

At $N = 25$, the residual models based on $P_1 = \{2\}$, $P_2 = \{2, 3\}$, and $P_3 = \{2, 3, 5\}$ have the exact data

P_r	# states	B_{25}	#RHC	$\#\mathrm{RHF}_{25,0.10}$
P_1	$3^{11} = 177147$	7	1296	162
P_2	$3^6 = 729$	7	18	3
P_3	1	2	0	0.

For P_2 the endpoint can be written

$$S_{25} = 2 + z_{13} + z_{17} + z_{19} + z_{23} + z_{25},$$

while the core z_{11} is silent in the endpoint sum.

In general the residual endpoint has the form

$$S_N = A_r(N) + \sum_c w_{r,N}(c) z_c,$$

where

$$w_{r,N}(c) = \sum_{\substack{d \mid P_r^{\#} \\ dc \le N}} \mu(d)$$

over the squarefree products of the known primes that are allowed by the closure rule. Consequently

$$B_r^E(N) = |A_r(N)| + \sum_c |w_{r,N}(c)|.$$

The exact endpoint distribution is encoded by

$$x^{A_r(N)} \prod_c \bigl(x^{-w_c} + 1 + x^{w_c}\bigr).$$

The residual model has its own singleton obstruction. Every unresolved core

$$c > \max(N/2, p_r^2), \qquad (c, P_r^{\#}) = 1,$$

appears at only one endpoint coordinate. Therefore

$$B_{N,P_r} \ge L_r(N) := \#\{\max(N/2, p_r^2) < c \le N : (c, P_r^{\#}) = 1\}.$$

This obstruction disappears once $p_r^2 \ge N$, when the residual rule determines the whole prefix.

33 A global Möbius identity that is terminal in the prime-sign model

For a squarefree-supported multiplicative function f with $f(p) = \sigma_p \in \{-1, +1\}$, consider the exact identity

$$\sum_{d \le N} f(d) \left\lfloor \frac{N}{d} \right\rfloor = 1.$$

Let $g = \mathbf{1} * f$. Then

$$g(n) = \prod_{p \mid n} (1 + \sigma_p) \ge 0,$$

and the left-hand side is

$$\sum_{n \le N} g(n).$$

Since $g(1) = 1$, equality to one forces $g(n) = 0$ for every $2 \le n \le N$. In particular, for every prime $p \le N$,

$$1 + \sigma_p = 0,$$

so

$$\sigma_p = -1.$$

Thus this single exact global identity determines every prime sign up to N in this model. It is a terminal constraint rather than a gradual intermediate layer.

34 Explicit rule systems and locality barriers

A completion set must be defined relative to a declared rule system. Write

$$\mathcal{C}_{N,R}(\mathcal{I})$$

for the words satisfying the information $\mathcal{I}$ after closure under the explicit rule set R. Full logical closure under every true fact about Möbius is not a usable intermediate model because Möbius is computable and sufficiently strong identities can recover the complete word.

A useful locality test treats linear constraints as rows of a matrix A. Let U be a target set of coordinates and let

$$d_U = \max_j |\operatorname{supp}(A_j) \cap U|.$$

If a dual combination uses only s rows, then

$$\left\|\mathbf{1}_U - (A^T y)_U\right\|_1 \geq |U| - s d_U.$$

The reason is that outside the union of the selected row supports, $(A^T y)_n = 0$, so the residual against $\mathbf{1}_U$ is exactly one.

For divisor-local rows and the upper-half squarefree set

$$U_N = \{n : N/2 < n \leq N,\ n \text{ squarefree}\},$$

each row touches at most one coordinate of U_N. Hence $d_U = 1$ and

$$|U_N| = Q(N) - Q(N/2) \sim \frac{3}{\pi^2} N.$$

Any dual certificate with $o(N)$ residual therefore needs $(3/\pi^2 - o(1))N$ such rows. This is a raw-coordinate divisor-incidence statement. Multiplicativity compresses many composite coordinates into prime variables, so it has the different $N/\log N$ singleton obstruction described above.

For pointwise translated windows of bounded length h, one has $d_U = O(h)$ and the same argument requires $\Omega(N/h)$ individual rows to cover a positive fraction of the target set. Aggregated translated statistics can touch $\Theta(N)$ positions and are not covered by this simple barrier. Global rows such as n^{-it} or $e(n\alpha)$ also touch essentially all coordinates.

For the linear relaxation

$$B_{N,+}^{\mathrm{LP}} = \max\{\mathbf{1}^T x : Ax = b,\ |x_n| \leq 1\},$$

linear-programming duality gives

$$B_{N,+}^{\mathrm{LP}} = \min_y \left(b^T y + \|\mathbf{1} - A^T y\|_1\right).$$

The absolute endpoint envelope is obtained by applying the same formula to both signs of the objective. This is a certificate language for linear models. Multiplicativity is nonlinear in the raw coordinates and must be lifted or parametrized before this formula can represent it. In the simple residual endpoint model $S = A + \sum_c w_c z_c$ with $z_c \in [-1, 1]$, the relaxation has zero integrality gap and reproduces $|A| + \sum_c |w_c|$.

For a hybrid system with g global rows $G_1, \ldots, G_g$ and s local rows each touching at most d coordinates of U, define

$$\eta_{g,s}(U) = \inf_\alpha \min_{\substack{T \subset U \\ |T| \leq sd}} \sum_{n \in U \setminus T} \left|1 - \sum_{j=1}^{g} \alpha_j G_j(n)\right|.$$

This is the remaining L^1 defect after the local rows are allowed to repair at most sd coordinates. It gives a common way to compare divisorial, translational, global, and hybrid certificate families.

35 Aggregated divisor identities and the inversion barrier

The Möbius identity

$$\sum_{d|n} \mu(d) = \mathbf{1}_{n=1}$$

is divisor-local for one value of n, but weighted aggregation produces a global row. Multiply by arbitrary real weights a_n and sum over $n \le N$. Then

$$\sum_{d\le N} \mu(d)c_d = a_1, \qquad c_d = \sum_{m\le N/d} a_{dm}.$$

Therefore

$$M(N) = a_1 + \sum_{d\le N} \mu(d)(1 - c_d),$$

and, because $\mu(d) = 0$ off the squarefree set,

$$|M(N)| \le |a_1| + \sum_{\substack{d\le N \\ d \text{ squarefree}}} |1 - c_d|.$$

This is an exact finite dual-style certificate for every choice of the weights a_n.

If one asks for the exact target $c_d = 1$ for every $d \le N$, inversion over the divisibility-by-multiples poset gives

$$a_n = \sum_{k\le N/n} \mu(k) = M(\lfloor N/n \rfloor).$$

In particular

$$a_1 = M(N).$$

Thus an exact one-row representation of the all-ones endpoint objective carries the unknown Mertens value in its right-hand side. The unrestricted variational problem is therefore exact but tautological:

$$|M(N)| = \inf_{a\in\mathbb{R}^N} \left(|a_1| + \sum_{\substack{d\le N \\ d \text{ squarefree}}} \left| 1 - \sum_{m\le N/d} a_{dm} \right| \right).$$

The research content can only come from restricting the weights to a natural sparse, local, smooth, arithmetic, or otherwise independently defined class.

36 Short correlations and the first-witness lag

For a finite word $x = (x_1, \dots, x_N)$, define

$$C_h(x; N) = \sum_{n\le N-h} x_n x_{n+h}, \qquad 1 \le h < N.$$

The exact identity

$$S_x(N)^2 = \sum_{n\le N} x_n^2 + 2\sum_{h=1}^{N-1} C_h(x; N)$$

holds for every real word. If x and the Möbius word have the same zero support, then the diagonal terms cancel and

$$\sum_{h=1}^{N-1}\big(C_h(x;N)-C_h(\mu;N)\big)=\frac{S_x(N)^2-M(N)^2}{2}.$$

Consequently, whenever $|S_x(N)|>|M(N)|$, at least one lag satisfies

$$C_h(x;N)>C_h(\mu;N).$$

The nontrivial question is how early such a witness must appear.

Define

$$h_*(x;N)=\min\{h\ge 1: C_h(x;N)>C_h(\mu;N)\},$$

with the value `NO_WITNESS` if the set is empty. For the excursion class $\mathrm{RHF}_{N,\delta}$ define

$$H_{\max,\delta}(N)=\max_{x\in\mathrm{RHF}_{N,\delta}} h_*(x;N),$$

with no fixed upper cutoff built into the definition.

The survivor curve is

$$R_{N,\delta}(H)=\#\{x\in\mathrm{RHF}_{N,\delta}: C_h(x;N)\le C_h(\mu;N)\text{ for all }h\le H\},$$

and the first-witness histogram is

$$W_{N,\delta}(h)=\#\{x\in\mathrm{RHF}_{N,\delta}: h_*(x;N)=h\}.$$

They satisfy

$$R_{N,\delta}(H)=R_{N,\delta}(H-1)-W_{N,\delta}(H).$$

The conditional elimination fraction at lag h is

$$q_{N,\delta}(h)=\frac{W_{N,\delta}(h)}{R_{N,\delta}(h-1)}.$$

High quantiles of h_* are recorded separately from the maximum because a small number of difficult states can control $H_{\max,\delta}$.

For a state with $S_x(N)^2>M(N)^2$, define the total correlation excess

$$\Delta_{\mathrm{total}}(x;N)=\frac{S_x(N)^2-M(N)^2}{2}$$

and the short-lag fraction

$$F_H(x;N)=\frac{\sum_{h=1}^{H}(C_h(x;N)-C_h(\mu;N))}{\Delta_{\mathrm{total}}(x;N)}.$$

This quantity need not be positive or monotone in H, but it measures how much of the endpoint excess has appeared by lag H.

An equivalent two-parameter view is to record, for each lag cutoff H, the largest effective power excess among states that still have no witness:

$$\Delta_{\max}(N,H)=\max_{x:\ h_*(x;N)>H}\left(\frac{\log|S_x(N)|}{\log N}-\frac{1}{2}\right).$$

This asks how large an excursion can remain hidden from the first H correlations.

37 Exact center of the fixed-σ_2 sign cube

Put the uniform probability measure on the $2^{\pi(N)-1}$ choices of the odd prime signs, with $\sigma_2 = -1$ fixed. For $N \geq 2$,

$$\mathbb{E}S_N = 0.$$

Indeed, $x_1 + x_2 = 0$, and every other nonzero squarefree term contains at least one free odd-prime sign of mean zero.

The correlation center can also be computed exactly. For two distinct squarefree integers n and $n + h$, the product $x_n x_{n+h}$ has nonzero expectation only when their sets of odd prime factors are equal. The only positive-lag possibility is

$$n = h, \qquad n + h = 2h,$$

with h odd and squarefree. Therefore

$$\mathbb{E}C_h = \begin{cases} -1, & h \text{ odd and squarefree and } 2h \leq N, \\ 0, & \text{otherwise.} \end{cases}$$

Thus the true Möbius word can be compared with an exact finite barycenter rather than with a Monte Carlo estimate.

For a lag cutoff H, write

$$\mathbf{C}_H(x) = (C_1(x), \ldots, C_H(x)), \qquad \overline{\mathbf{C}}_H = \mathbb{E}\mathbf{C}_H.$$

Let Σ_H be the exact covariance matrix over the finite sign cube. A standardized centrality statistic is

$$D_H(x)^2 = \big(\mathbf{C}_H(x) - \overline{\mathbf{C}}_H\big)^T \Sigma_H^{\dagger} \big(\mathbf{C}_H(x) - \overline{\mathbf{C}}_H\big),$$

where $\Sigma_H^{\dagger}$ is the Moore–Penrose pseudoinverse.

Centrality is only one possible geometry. A second test is Pareto order: the Möbius vector is Pareto-minimal up to lag H if there is no different admissible state x such that

$$C_h(x) \leq C_h(\mu) \qquad \text{for every } 1 \leq h \leq H.$$

A third test is excursion-separation performance. For a reference word r, define

$$R_{N,\delta,r}(H) = \#\{x \in \mathrm{RHF}_{N,\delta} : C_h(x) \leq C_h(r) \text{ for every } h \leq H\}$$

and, for a fixed scan depth H_{scan},

$$A_{N,\delta}(r) = \sum_{H=0}^{H_{\mathrm{scan}}} \frac{R_{N,\delta,r}(H)}{R_{N,\delta,r}(0)}.$$

Small $A_{N,\delta}(r)$ means that the reference eliminates excursion states quickly. These three tests—distance to the barycenter, Pareto order, and adversarial separation—are kept distinct.

38 Exact short-correlation benchmark at $N = 25$

In the fixed-σ_2 multiplicative family there are 256 states at $N = 25$. The true Möbius word has

$$S_{25} = -2$$

and

$$(C_1, \ldots, C_9) = (-3, -1, -5, 1, -1, 5, -2, 1, -1).$$

The exact barycenter of the first nine correlations is

$$(-1, 0, -1, 0, -1, 0, -1, 0, 0).$$

Using equality of successive correlations, the true Möbius word is selected from the 256 states by

$$256 \xrightarrow{C_1=-3} 56 \xrightarrow{C_2=-1} 19 \xrightarrow{C_3=-5} 1.$$

Thus the first three correlation values identify the true word uniquely inside this finite admissible family.

There is also an order statement. Among all 256 states, the true Möbius word is the unique state satisfying

$$C_1(x) \le -3, \qquad C_2(x) \le -1, \qquad C_3(x) \le -5.$$

Equivalently, it is already Pareto-minimal in the first three coordinates.

The one-step transition counts from a -1 entry in the true word are

$$N_{--} = 1, \qquad N_{-0} = 6, \qquad N_{-+} = 2.$$

These three sequential counts also identify the true word uniquely among the 256 states.

For the first nine correlations, the exact covariance matrix is nonsingular. The squared Mahalanobis distance of the Möbius vector from the barycenter is

$$D_9(\mu)^2 = 7.54861104228501,$$

so

$$D_9(\mu) = 2.747473574447079.$$

Exactly 104 of the 256 states have Mahalanobis distance at most this value, placing the Möbius word at the 40.625% empirical cumulative rank by this particular centrality statistic. Thus it is not the unique Euclidean-type center of the cloud. Its coordinate positions are strongly mixed: for $C_3 = -5$, only four states have a smaller value, while for $C_6 = 5$, only two states have a larger value. This is why coordinatewise averages, multivariate centrality, and Pareto balance must be tested separately.

The eight RHC states at $N = 25$ all have $|S| = 6$. For the true word, $C_2 = -1$, whereas every one of these eight RHC states has

$$C_2 \in \{3, 5, 7, 9\}.$$

Thus lag 2 alone separates the true word from every RHC state at this scale.

39 Discovery-stage larger-N correlation observations

An earlier exploratory sweep reported the following discovery-stage behavior in the same fixed-σ_2 family. For every tested $N \leq 106$, every RHC state was reported to have a positive correlation witness by lag 9, with the largest reported first-witness lag occurring at $N = 105$. The same exploratory work reported that at most nine exact short correlations were sufficient to identify the true Möbius word among all admissible states for every tested $N \leq 100$.

These figures are retained as discovery targets, not as final computational benchmarks. The current C++ implementation is being built with an independent reference engine specifically to revalidate the state-by-state correlation calculations and to continue beyond the discovery range without fixing 9 as a cutoff. The production quantity is the full function $H_{\max,\delta}(N)$, not the statement $H \leq 9$.

40 Null references and multiple-comparison control

A short-lag feature found after inspecting many states and many lags can occur by selection. The finite experiment therefore compares the true Möbius word with other reference words from the same admissible family.

For a reference r, compute $R_{N,\delta,r}(H)$ and $A_{N,\delta}(r)$ as above. Useful reference classes are:

1. uniform references from the full admissible cube;
2. uniform references conditioned to be outside the RHC class;
3. references conditioned on the exact endpoint $S_r = M(N)$;
4. if the exact endpoint fiber is too small, references with $|S_r - M(N)| \leq w$, with the width w and the fiber size reported explicitly.

The endpoint-matched class is especially important because it removes the simplest explanation that a short-correlation signature is caused only by the small value of the endpoint sum.

For a chosen statistic $T(r)$, such as $A_{N,\delta}(r)$, $H_{\max,\delta}(r)$, a Mahalanobis centrality score, or a Pareto rank, the exact finite rank of the true Möbius word can be computed whenever all references are enumerable. Sampling is used only when the reference space is too large. The discovery range and the validation range are kept separate when a statistic or cutoff was selected after inspecting the discovery data.

41 Bridge from short correlations to forbidden Möbius excursions

The target implication for fixed $\delta > 0$ is a deterministic statement of the form

$$|S_x(N)| > N^{1/2+\delta} \quad \Longrightarrow \quad \exists h \leq H_\delta(N) : C_h(x; N) > C_h(\mu; N)$$

for every admissible state x in the declared family. If this implication is proved for all sufficiently large N, then it can be applied to the state $x = \mu$. The conclusion would require

$$C_h(\mu; N) > C_h(\mu; N),$$

which is impossible. Therefore the premise cannot hold for $x = \mu$, and

$$|M(N)| \leq N^{1/2+\delta}$$

for all sufficiently large N in the theorem's range. If such a statement is proved for every fixed $\delta > 0$, it gives

$$M(N) = O_\varepsilon(N^{1/2+\varepsilon})$$

and hence RH.

The case $\delta = 0$ cannot satisfy an analogous universal theorem for all N, because that would imply the disproved Mertens bound $|M(N)| \le \sqrt{N}$. For this reason the computational program tracks $\delta = 0$ as a baseline and several positive values of δ simultaneously. The useful asymptotic outcome would be a clear separation in which the square-root case deteriorates while every fixed positive excess remains localized to a much shorter lag scale.

42 Canonical finite benchmark for $1 \le N \le 25$

This section fixes a single reproducible finite benchmark. All counts are exact. The unrestricted ternary layer is computed by coefficient convolution of $(z^{-1} + 1 + z)^N$; the squarefree-support layer fixes $\mu(1) = 1$, fixes zero on nonsquarefree indices, and leaves the remaining squarefree signs free; the multiplicative layer assigns free signs to the primes $p \le N$ and extends multiplicatively on squarefree integers; the last nonterminal layer additionally fixes only $\mu(2) = -1$. No true values $M(N)$ are used in these four layers.

The strict endpoint classes used here are

$$\mathrm{RHC}_N = \{\omega : |S_\omega(N)| > \sqrt{N}\}, \qquad \mathrm{RHF}_{N,\delta} = \{\omega : |S_\omega(N)| > N^{1/2+\delta}\}.$$

The original reference population 3^N is retained in every table. A terminal control layer obtained by imposing the full prime rule $\mu(p) = -1$ for every prime $p \le N$ contains exactly one word and is therefore excluded from the nontrivial filtration.

42.1 Surviving completion counts

N	3^N	squarefree support	multiplicative	multiplicative $+\mu(2) = -1$
1	3	1	1	1
2	9	2	2	1
3	27	4	4	2
4	81	4	4	2
5	243	8	8	4
6	729	16	8	4
7	2,187	32	16	8
8	6,561	32	16	8
9	19,683	32	16	8
10	59,049	64	16	8
11	177,147	128	32	16
12	531,441	128	32	16
13	1,594,323	256	64	32
14	4,782,969	512	64	32
15	14,348,907	1,024	64	32
16	43,046,721	1,024	64	32
17	129,140,163	2,048	128	64
18	387,420,489	2,048	128	64
19	1,162,261,467	4,096	256	128
20	3,486,784,401	4,096	256	128
21	10,460,353,203	8,192	256	128
22	31,381,059,609	16,384	256	128
23	94,143,178,827	32,768	512	256
24	282,429,536,481	32,768	512	256

N	3^N	squarefree support	multiplicative	multiplicative $+\mu(2) = -1$
25	847,288,609,443	32,768	512	256

42.2 Extremal envelope and archival finite effective exponent

	ternary		squarefree		multiplicative		$+\mu(2) = -1$	
N	B_N	E_N	B_N	E_N	B_N	E_N	B_N	E_N
1	1	–	1	–	1	–	1	–
2	2	+0.50000	2	+0.50000	2	+0.50000	0	–
3	3	+0.50000	3	+0.50000	3	+0.50000	1	-0.50000
4	4	+0.50000	3	+0.29248	3	+0.29248	1	-0.50000
5	5	+0.50000	4	+0.36135	4	+0.36135	2	-0.06932
6	6	+0.50000	5	+0.39824	5	+0.39824	1	-0.50000
7	7	+0.50000	6	+0.42078	6	+0.42078	2	-0.14379
8	8	+0.50000	6	+0.36165	6	+0.36165	2	-0.16667
9	9	+0.50000	6	+0.31546	6	+0.31546	2	-0.18454
10	10	+0.50000	7	+0.34510	7	+0.34510	1	-0.50000
11	11	+0.50000	8	+0.36719	8	+0.36719	2	-0.21094
12	12	+0.50000	8	+0.33683	8	+0.33683	2	-0.22106
13	13	+0.50000	9	+0.35663	9	+0.35663	3	-0.07168
14	14	+0.50000	10	+0.37250	10	+0.37250	2	-0.23735
15	15	+0.50000	11	+0.38547	11	+0.38547	3	-0.09432
16	16	+0.50000	11	+0.36486	11	+0.36486	3	-0.10376
17	17	+0.50000	12	+0.37706	12	+0.37706	4	-0.01070
18	18	+0.50000	12	+0.35972	12	+0.35972	4	-0.02038
19	19	+0.50000	13	+0.37112	13	+0.37112	5	+0.04660
20	20	+0.50000	13	+0.35620	13	+0.35620	5	+0.03724
21	21	+0.50000	14	+0.36682	14	+0.36682	6	+0.08852
22	22	+0.50000	15	+0.37610	15	+0.37610	5	+0.02068
23	23	+0.50000	16	+0.38426	16	+0.38426	6	+0.07144
24	24	+0.50000	16	+0.37242	16	+0.37242	6	+0.06379
25	25	+0.50000	16	+0.36135	16	+0.36135	6	+0.05664

42.3 RHC survivor counts

N	3^N	ternary RHC	squarefree RHC	multiplicative RHC	$+\mu(2)$ RHC
1	3	0	0	0	0
2	9	2	1	1	0
3	27	8	1	1	0
4	81	10	1	1	0
5	243	42	1	1	0
6	729	156	6	2	0
7	2,187	548	7	3	0
8	6,561	1,854	7	3	0
9	19,683	3,014	7	3	0
10	59,049	10,386	8	2	0
11	177,147	34,938	37	7	0
12	531,441	115,440	37	7	0
13	1,594,323	376,196	46	10	0
14	4,782,969	1,212,672	176	16	0
15	14,348,907	3,874,980	232	10	0
16	43,046,721	7,253,118	232	10	0
17	129,140,163	23,469,114	299	15	0
18	387,420,489	75,346,590	299	15	0
19	1,162,261,467	240,301,818	1,093	53	8
20	3,486,784,401	762,080,202	1,093	53	8

N	3^N	ternary RHC	squarefree RHC	multiplicative RHC	$+\mu(2)$ RHC
21	10,460,353,203	2,405,119,464	1,471	33	4
22	31,381,059,609	7,558,657,992	4,944	56	8
23	94,143,178,827	23,667,691,934	6,885	82	8
24	282,429,536,481	73,869,142,280	6,885	82	8
25	847,288,609,443	150,937,165,552	6,885	82	8

42.4 $\mathbf{RHF}_{0.05}$ survivor counts

N	3^N	ternary	squarefree	multiplicative	$+\mu(2)$
1	3	0	0	0	0
2	9	2	1	1	0
3	27	8	1	1	0
4	81	10	1	1	0
5	243	42	1	1	0
6	729	156	6	2	0
7	2,187	548	7	3	0
8	6,561	846	7	3	0
9	19,683	3,014	7	3	0
10	59,049	10,386	8	2	0
11	177,147	34,938	37	7	0
12	531,441	115,440	37	7	0
13	1,594,323	200,592	46	10	0
14	4,782,969	672,132	56	8	0
15	14,348,907	2,220,600	232	10	0
16	43,046,721	7,253,118	232	10	0
17	129,140,163	23,469,114	299	15	0
18	387,420,489	75,346,590	299	15	0
19	1,162,261,467	141,568,470	378	23	0
20	3,486,784,401	459,240,570	378	23	0
21	10,460,353,203	1,479,280,662	1,471	33	4
22	31,381,059,609	4,736,045,276	1,941	26	0
23	94,143,178,827	15,082,716,810	6,885	82	8
24	282,429,536,481	47,810,887,112	6,885	82	8
25	847,288,609,443	150,937,165,552	6,885	82	8

42.5 $\mathbf{RHF}_{0.10}$ survivor counts

N	3^N	ternary	squarefree	multiplicative	$+\mu(2)$
1	3	0	0	0	0
2	9	2	1	1	0
3	27	8	1	1	0
4	81	10	1	1	0
5	243	42	1	1	0
6	729	156	6	2	0
7	2,187	226	7	3	0
8	6,561	846	7	3	0
9	19,683	3,014	7	3	0
10	59,049	10,386	8	2	0
11	177,147	16,854	9	3	0
12	531,441	58,812	9	3	0
13	1,594,323	200,592	46	10	0
14	4,782,969	672,132	56	8	0
15	14,348,907	1,157,538	67	5	0
16	43,046,721	3,922,206	67	5	0
17	129,140,163	13,105,098	299	15	0

N	3^N	ternary	squarefree	multiplicative	$+\mu(2)$
18	387,420,489	43,282,686	299	15	0
19	1,162,261,467	141,568,470	378	23	0
20	3,486,784,401	257,959,890	378	23	0
21	10,460,353,203	851,645,150	470	17	0
22	31,381,059,609	2,788,013,542	1,941	26	0
23	94,143,178,827	9,060,478,368	2,517	40	0
24	282,429,536,481	29,257,136,160	2,517	40	0
25	847,288,609,443	93,945,019,352	2,517	40	0

42.6 $\mathrm{RHF}_{0.20}$ survivor counts

N	3^N	ternary	squarefree	multiplicative	$+\mu(2)$
1	3	0	0	0	0
2	9	2	1	1	0
3	27	2	1	1	0
4	81	10	1	1	0
5	243	12	1	1	0
6	729	56	1	1	0
7	2,187	226	7	3	0
8	6,561	314	1	1	0
9	19,683	1,250	1	1	0
10	59,049	1,782	1	1	0
11	177,147	7,020	9	3	0
12	531,441	26,340	9	3	0
13	1,594,323	39,860	10	4	0
14	4,782,969	149,610	11	3	0
15	14,348,907	544,068	67	5	0
16	43,046,721	1,929,774	67	5	0
17	129,140,163	3,125,622	79	8	0
18	387,420,489	11,148,402	79	8	0
19	1,162,261,467	39,021,138	92	12	0
20	3,486,784,401	64,683,150	92	12	0
21	10,460,353,203	227,686,670	106	11	0
22	31,381,059,609	789,507,890	576	14	0
23	94,143,178,827	2,702,576,730	697	22	0
24	282,429,536,481	4,644,553,260	697	22	0
25	847,288,609,443	15,993,177,852	697	22	0

42.7 Immediate finite observations

The benchmark separates reduction in state count from reduction in the extremal envelope. The E_N columns are retained only as exact finite diagnostics; the closed forms above show that all four baseline envelopes have exponent one. At $N = 25$ the squarefree-support layer has 32768 words and the multiplicative layer has only 512, yet both have $B_{25} = 16$. Fixing $\sigma_2 = -1$ changes the constant sharply, from 16 to 6, but the closed form above shows that this layer still has linear asymptotic growth. At $N = 25$ it leaves eight RHC states and no $\mathrm{RHF}_{25,0.10}$ state. The short-correlation benchmark adds a different kind of information: the first three correlations identify the true word uniquely among the 256 fixed-σ_2 states, and lag 2 separates it from all eight RHC states.

43 Conclusion

The first baseline computation is now analytically closed: $B_N^{\mathrm{sqf}} = B_N^{\mathrm{mult}} = Q(N)$ and $B_N^{\mu(2)} = Q_{\mathrm{odd}}(N) - Q_{\mathrm{even}}(N)$. Hence the apparent finite drop $16 \to 6$ is an asymptotic change of constant from $6/\pi^2$ to $2/\pi^2$, not a change of power, and the finite quantity E_N is unsuitable as the main scaling diagnostic. The exact finite tables remain valid, while B_N/N and local log–log slopes provide the correct scaling test.

The arithmetic-completion and short-correlation programs now have separate exact objects. The first asks how declared Möbius information changes an adversarial endpoint envelope. The second fixes a finite admissible multiplicative cube and asks how quickly a large endpoint excursion must depart from the true Möbius correlation vector. The baseline arithmetic layers with squarefree support, multiplicativity, and only $\sigma_2 = -1$ have linear extremal envelopes, so their small finite values do not constitute a power contraction. Raw prime revelation and residual closure have different singleton obstructions and must remain separate models. Strict divisor-local constraints have a support-coverage barrier, while aggregated divisor identities can create global rows but recover the Mertens value in the exact all-ones representation.

For short correlations, the endpoint identity forces some positive total correlation excess whenever $|S_x| > |M(N)|$, but it does not explain why the witness should occur at a small lag. The new computational target is therefore the growth of $H_{\max,\delta}(N)$ and the geometry of the hardest surviving states. The finite sign cube has an exact barycenter and covariance structure, which allows the true Möbius word to be tested for ordinary centrality, Pareto extremality, and excursion-separation performance without choosing one interpretation in advance. At $N = 25$, the true word is not an ordinary Mahalanobis outlier but is uniquely Pareto-minimal in the first three correlations and is uniquely identified by those same three correlation values.

The surrounding literature provides close models—random multiplicative functions, biased random multiplicative functions, sign-pattern results, discrepancy, pretentious distance, inverse theorems, and rigidity results—but the finite correlation position of the true Möbius word inside the full admissible prime-sign cube is kept as a separate explicitly defined object. The next stage is exact revalidation and scaling of the C++ exhaustive engine, followed by analytic work on any stable sublinear witness law that survives the larger computations.

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